2. key principle of ACM code.

1. General Ethics principle
2. Professional Responsibilities.
3. Professional leadership principle
4. compliance with the code.

9. A data scientist building a pandemic tracking tool may face a conflict between contributing to society and respecting privacy. The ACM code guides resolution by prioritizing the public Interest, the minimum necessary data collection and robust security to mitigate harm to individual privacy.

10.

copy rights	patent	Trade mark
creative expression	functional Invention	Brand Identity
life + 70 years	20 years	indefinite
source code	methods	Trade secret

3. The 8 core pillars

1. Public
2. client and Employer
3. Product
4. Judgement
5. management
6. Profession
7. collegues
8. self

Key Professional Issues.

1. safety vs speed.
2. whistleblowing
3. Privacy vs profit

4. Utilitarianism - focuses on consequences. choosing action that creates the greatest good for the g nembers. In a privacy dilleman, a Utilitarian justify data collection if it helps tracks a pndem stop a crime. Prioritizing social benefits over individual privacy.

02. 1. Therac-25: In 1980s, a radiation therapy machine delivered lethal doses to patients due to software "race condition".

lesson: software can kill.

2. Volkswagen "Dieselgate" (Intentional fraud)

Engineers wrote code specifically designed to detect when a car was being tested for emissions. The car would "cheat" by lowering its pollution only during the test, then emitting 40 times the legal limit on the road.

lesson: Just following orders is not a defense.

[illegible]

4 Explainability ensures accountability and fairness by allowing humans to understand AI decisions. This legal basis is important especially in life altering situations without it, biases may go unchecked leading to systematic discrimination and loss of trust in AI systems.

11 open source

fully accessable

free

Transparent

proprietary

secret

subscription

black box

13 Ethical principle

Privacy by design

User autonomy

Normative fairness

The goal

Building privacy into the code from day one.

Giving the users total control over their digital footprint

Ensuring data collection does not cause "algorithmic harm"

14 Principle

1. Data minization
2. Purpose limitation
3. Transparency
4. Security & Integrity
5. User autonomy